PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Gaurav Ravindra Chalke
Program/Batch	B084-Networkwalks
Date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks.com 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** Not performed; outside the selected Week 2 modules.

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report documents the cybersecurity practical activities completed during NetworkWalks Week 2. The project focused on reconnaissance, footprinting and network discovery using Kali Linux. Two modules were selected and performed:

1. W2-PM1 – Footprinting with Multiple Kali Tools.
2. W2-PM5 – Zenmap Based Network Scanning.

The purpose was to collect domain-related information using reconnaissance tools, identify network configuration details, discover live hosts and generate a network topology.

Screenshots and command outputs were collected to document the practical activities and their results.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance on **networkwalks.com** using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon.

- **WHOIS:** Collected domain registration information and name server details.
- **WhatWeb:** Identified website technologies, including WordPress 7.0.4 and WP Download Manager 3.3.58.
- **Nslookup:** Resolved the domain to IP address 192.232.216.135.

- **Curl:** Examined HTTP response headers and identified the `/wp-json/` REST API endpoint.
- **Wafw00f:** Detected ModSecurity (SpiderLabs) as the Web Application Firewall.
- **DNSRecon:** Enumerated DNS records, including name servers, mail servers, SPF/TXT and service records.

These activities helped collect publicly available information about the target website and its infrastructure.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** in Kali Linux to perform network discovery on my local network. The objective was to identify the local network range, discover active hosts, and generate a network topology.

Network Configuration:

- Operating System: Kali Linux
- Network Interface: `eth0`
- Local IP Address: `10.0.0.2`
- Network Range: `10.0.0.0/24`
- Subnet Mask: `255.255.255.0`
- Gateway: `10.0.0.1`

Scanning Process:

1. I checked the network configuration in Kali Linux using the terminal.
2. I opened Zenmap and entered the target network range `10.0.0.0/24`.
3. I selected **Ping Scan** to discover active devices on the local network.
4. I reviewed the scan results to identify the live hosts.
5. I opened the **Topology** tab, enabled the legend, and generated the network topology.

Scan Results:

The Ping Scan identified the following two active hosts:

- `10.0.0.1`
- `10.0.0.2` (Kali Linux machine)

The MAC address of my Kali Linux machine was identified as `08:00:27:5a:8f:2f`.

Network Topology:

After completing the scan, I viewed the discovered network devices in Zenmap's **Topology** section. I saved the generated topology as `W2-PM5-Network-Topology.png` and converted it into PDF format as `W2-PM5-Network-Topology.pdf`.

This activity helped me understand how to discover active devices on a local network and visualize the network structure using Zenmap.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	May help attackers identify software versions that require further security review.	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the web server's network location.	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further information gathering.	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the website's security infrastructure.	Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	Helps build a profile of the domain's infrastructure.	Low
6	Multiple live hosts visible on local network	Zenmap Ping Scan identified two live hosts: 10.0.0.1 and 10.0.0.2	Helps identify active devices on the local network.	Informational

Risk Level Key: **Critical** – High impact; **Medium** – Requires review; **Low** – Limited potential impact; **Informational** – Observation only.

The findings above are observations from the footprinting and network scanning exercises, not confirmed vulnerabilities. The practical activities focused on information gathering and host discovery. No exploitation or vulnerability validation was performed.

The presence of software version information, an identifiable IP address, DNS records or live network hosts does not automatically indicate a security weakness. Further authorized security testing would be required to confirm any actual vulnerabilities.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review Exposed Technology:** Regularly check publicly available information about website technologies, CMS and plugins.
2. **Keep Software Updated:** Apply security patches and updates to CMS platforms, plugins and web technologies.
3. **Review HTTP Headers:** Check response headers and limit unnecessary technical information disclosure.
4. **Review DNS Records:** Periodically verify DNS records and remove unnecessary public entries.
5. **Configure and Monitor WAF:** Keep ModSecurity properly configured, updated and monitored.
6. **Perform Network Discovery:** Regularly scan authorized internal networks to identify active devices.
7. **Investigate Unknown Devices:** Verify unexpected devices discovered during network scans.
8. **Maintain Network Documentation:** Keep network topology and device information updated.
9. **Perform Authorized Testing:** Conduct reconnaissance and security testing only with proper authorization.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

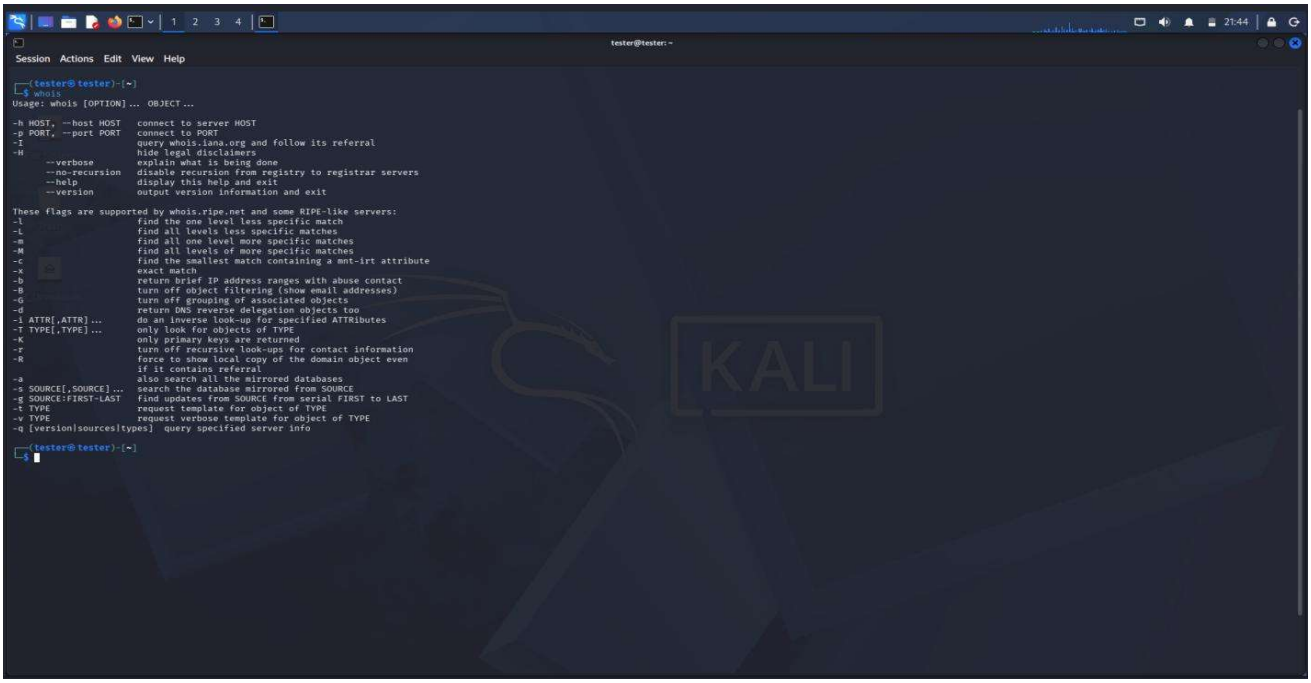
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I identified the active hosts' IP addresses and the MAC address of my Kali Linux machine.

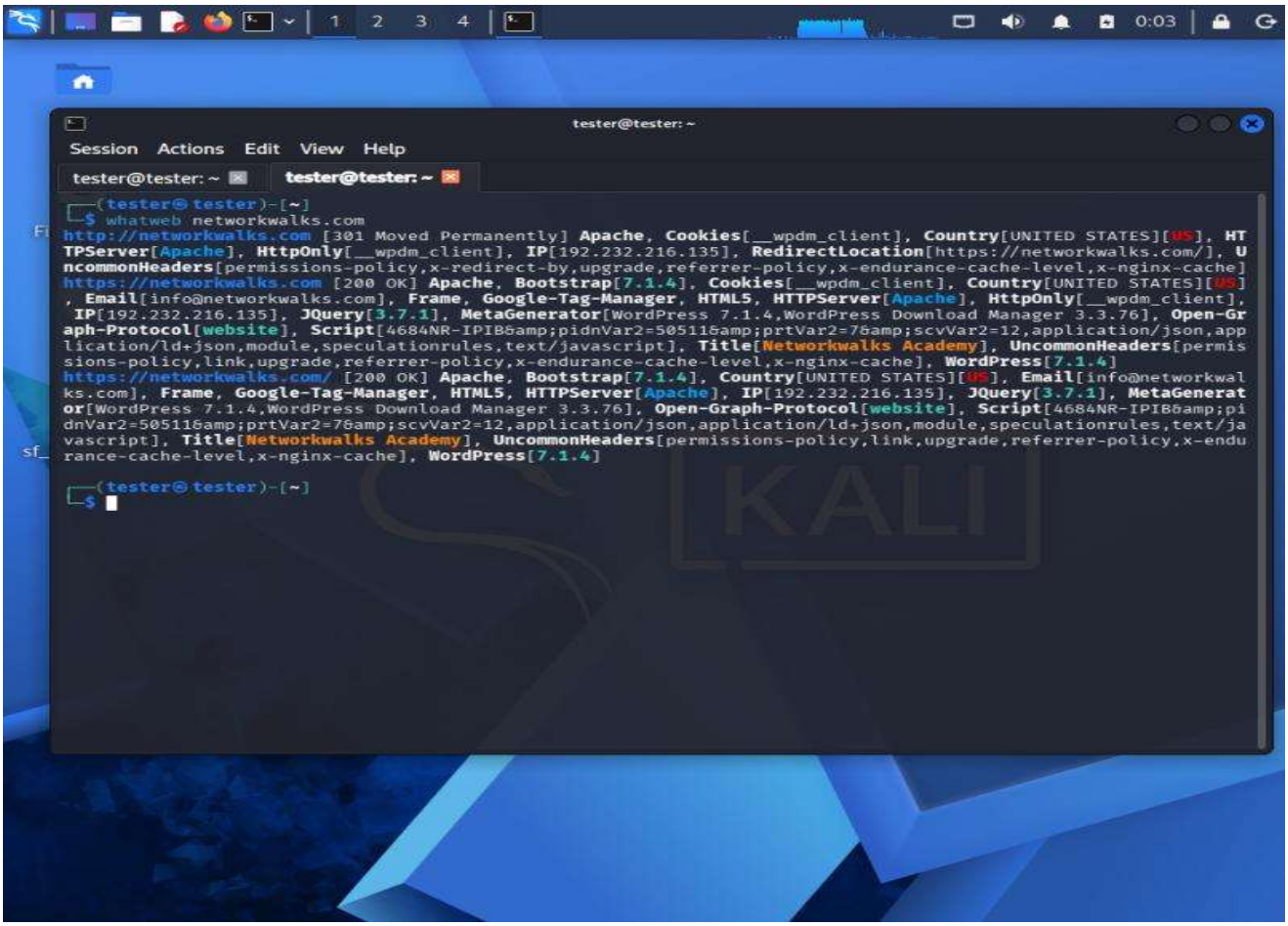
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

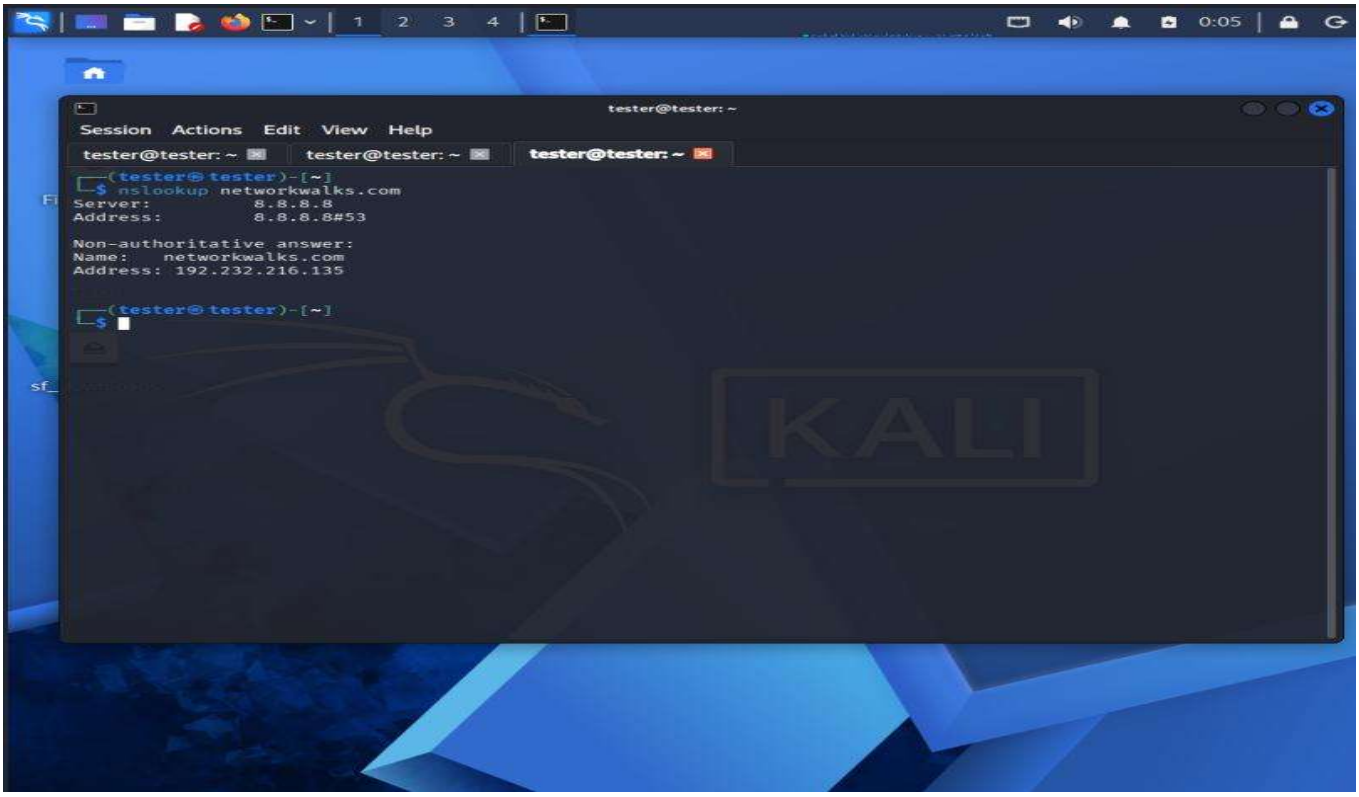
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

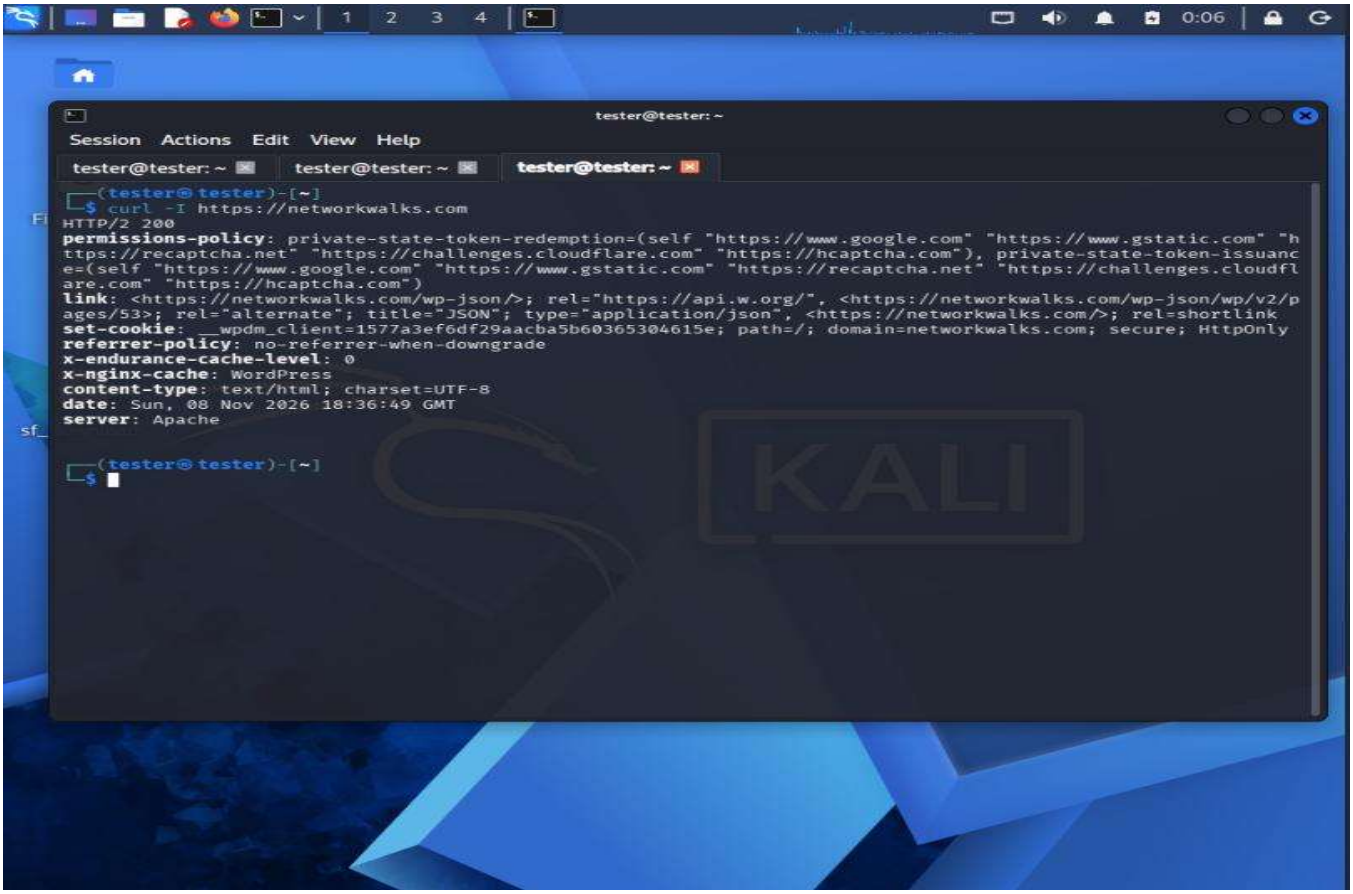
Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

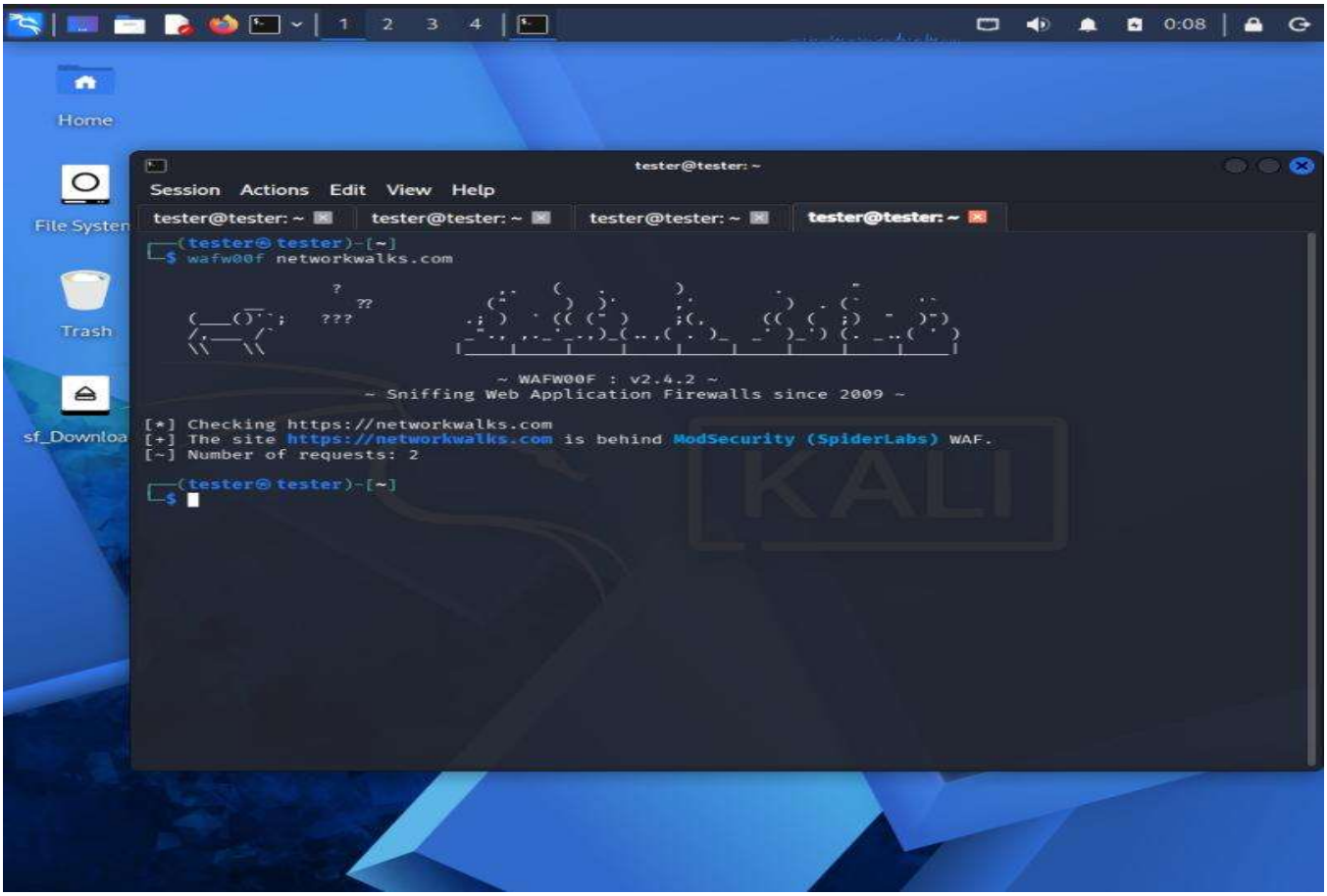
8. Evidences Collected

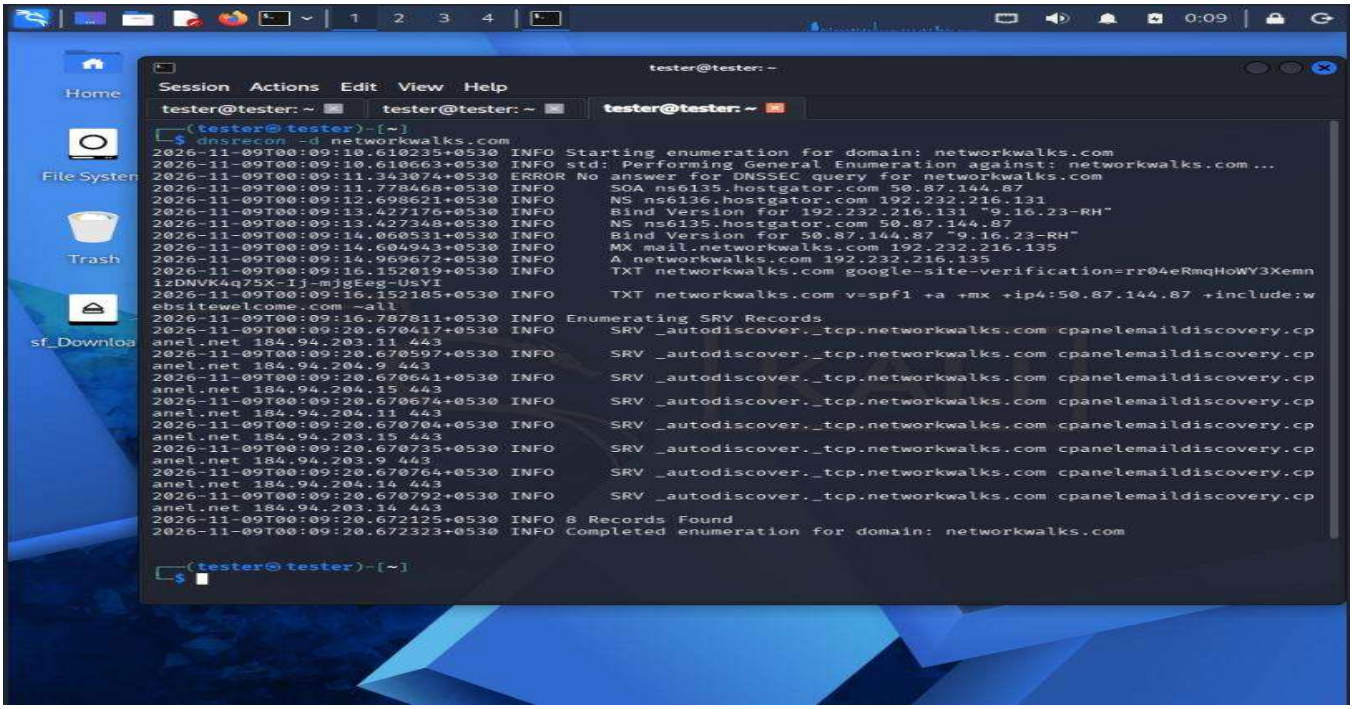


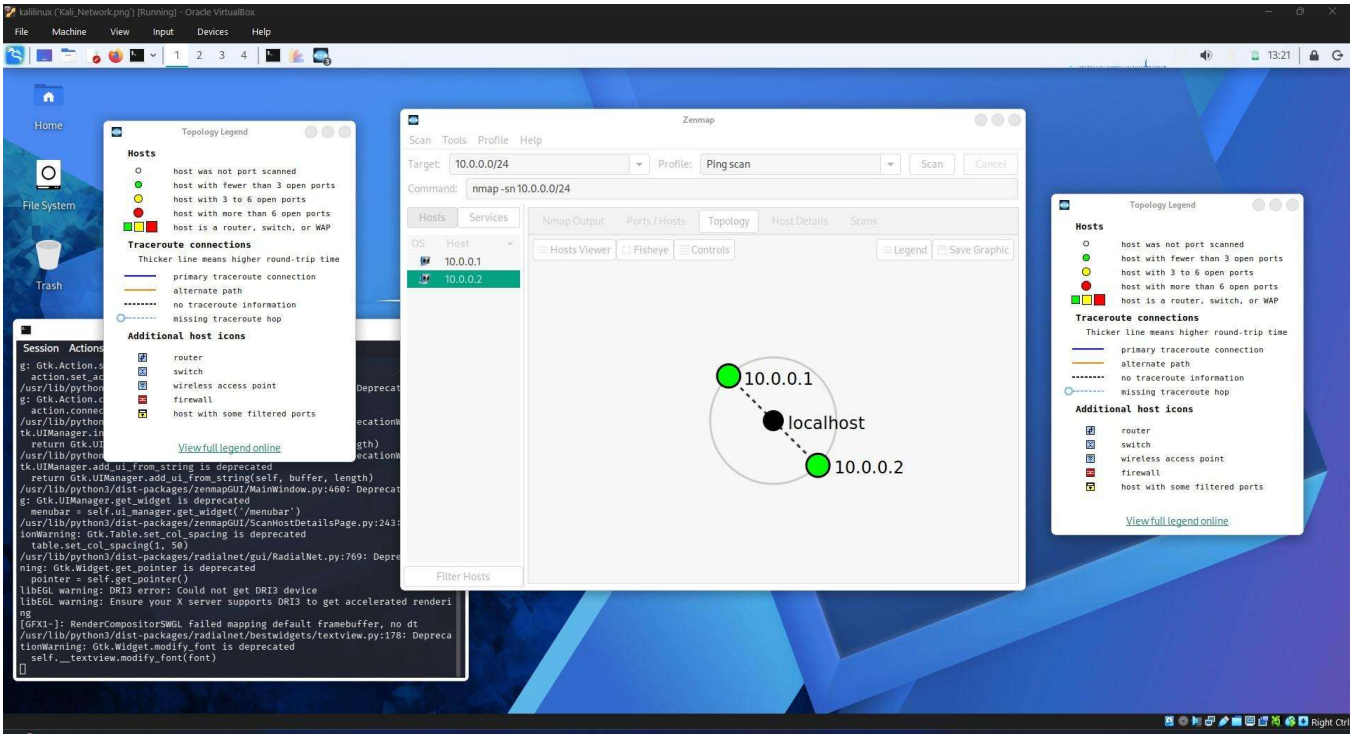



tester@tester: ~
Session Actions Edit View Help
tester@tester: ~
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tester@tester: ~
(tester㉿tester)-[~]
$ nslookup networkwalks.com
Server: 8.8.8.8
Address: 8.8.8.8#53
Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135
(tester㉿tester)-[~]
$


tester@tester: ~
Session Actions Edit View Help
tester@tester: ~
tester@tester: ~
tester@tester: ~
(tester㉿tester)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=1577a3ef6df29aacba5b60365304615e; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Sun, 08 Nov 2026 18:36:49 GMT
server: Apache
(tester㉿tester)-[~]
$


tester@tester: ~
Session Actions Edit View Help
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tester@tester: ~
tester@tester: ~
(tester㉿tester)-[~]
$ wafw00f networkwalks.com
~ WAFW00F : v2.4.2 ~
~ Sniffing Web Application Firewalls since 2009 ~
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
(tester㉿tester)-[~]
$


tester@tester: ~
Session Actions Edit View Help
tester@tester: ~
tester@tester: ~
tester@tester: ~
(tester㉿tester)-[~]
$ dnsrecon -d networkwalks.com
2026-11-09T00:09:10.610235+0530 INFO Starting enumeration for domain: networkwalks.com
2026-11-09T00:09:10.610663+0530 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-09T00:09:11.343074+0530 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-09T00:09:11.778468+0530 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-09T00:09:12.698621+0530 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-09T00:09:13.427176+0530 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-09T00:09:13.427348+0530 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-09T00:09:14.060531+0530 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-09T00:09:14.604943+0530 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-09T00:09:14.969672+0530 INFO A networkwalks.com 192.232.216.135
2026-11-09T00:09:16.152019+0530 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3Xemn
izDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-09T00:09:16.152185+0530 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:w
ebs1tewelcome.com ~all
2026-11-09T00:09:16.787811+0530 INFO Enumerating SRV Records
2026-11-09T00:09:20.670417+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.11 443
2026-11-09T00:09:20.670597+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.9 443
2026-11-09T00:09:20.670641+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.15 443
2026-11-09T00:09:20.670674+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.11 443
2026-11-09T00:09:20.670704+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.15 443
2026-11-09T00:09:20.670735+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.9 443
2026-11-09T00:09:20.670764+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.14 443
2026-11-09T00:09:20.670792+0530 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.14 443
2026-11-09T00:09:20.672125+0530 INFO 8 Records Found
2026-11-09T00:09:20.672323+0530 INFO Completed enumeration for domain: networkwalks.com
(tester㉿tester)-[~]
$



-End-

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub